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Inventor(s)

Brady; Mason

PROSTHETIC LIMB WITH THERMO-ELECTRIC ACTUATOR

Abstract

A prosthetic apparatus can include: a limb connection portion attachable to a user; a hand portion opposite the limb connection portion, the hand portion including digits; and thermo-electric actuators positioned between the hand portion and the limb connection portion, the thermo-electric actuators connected to the digits.

Inventors: Brady; Mason (Durango, CO)

Applicant: Brady; Mason (Durango, CO)

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Background/Summary

FIELD

[0001] The described examples relate generally to prosthetic limbs. In particular examples, the disclosure relates to prosthetic limbs having a thermo-electric actuator.

BACKGROUND

[0002] Prosthetics can include functional, artificial replacement limbs for users with amputated, malformed, or missing limbs. Prosthetics can often provide significant transformations to the quality of life for such users. In recent years, prosthetics have undergone great advancements with respect to their functionality and user adaptability. Unfortunately, however, some prosthetics—particularly prosthetic arms—can be too heavy for certain users. Users are often keenly aware of the weight of a prosthetic arm because the center of mass for the prosthetic arm is positioned so far away from the shoulder joint. Adding other components to a prosthetic arm, such as a battery, can add to the weight of the prosthetic arm and exacerbate the perceived effects. Accordingly, at least some conventional prosthetics are too heavy and unwieldy for users, lending to fatigue and lack of fine motor control. Some users can even develop injuries from repetitive compensatory movement patterns due at least in part to the heavy weight of a prosthetic. Therefore, there is a need for an improved prosthetic limb that is lighter and more user friendly.

[0003] The subject matter claimed herein is not limited to examples that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some examples described herein may be practiced.

SUMMARY

[0004] An aspect of the present disclosure relates to a prosthetic apparatus that includes: a limb connection portion attachable to a user; a hand portion opposite the limb connection portion, the hand comprising a plurality of digits; and a plurality of thermo-electric actuators positioned between the hand portion and the limb connection portion, the plurality of thermo-electric actuators connected to the plurality of digits.

[0005] The prosthetic apparatus can further include a plurality of cables connecting the plurality of thermo-electric actuators to the plurality of digits, wherein upon thermal activation of a thermo-electric actuator of the plurality of thermo-electric actuators, the thermo-electric actuator retracts a cable of the plurality of cables to move a digit of the plurality of digits inward to a retracted position. In some examples, a magnitude of digit retraction to the retracted position is dependent upon a level of the thermal activation. In certain examples, the thermo-electric actuator of the plurality of thermo-electric actuators is configured to transition from an activated state to a resting state by cooling down from a first temperature above an activation temperature to a second temperature below the activation temperature. In at least one example, upon transitioning the thermo-electric actuator of the plurality of thermo-electric actuators from the activated state to the resting state, the digit of the plurality of digits is configured to extend outward and away from the retracted position to a resting position.

[0006] The prosthetic apparatus can further include a biasing structure to bias the digit of the plurality of digits toward the resting position. In some examples a first end of the biasing structure is connected to a top portion of the digit between a mid-finger knuckle joint and a fingertip; and a second end of the biasing structure is connected to a top portion of the hand portion between a base knuckle joint and a wrist joint. In one or more examples, each thermo-electric actuator of the plurality of thermo-electric actuators includes at least one Nitinol strand. In particular examples, each thermo-electric actuator of the plurality of thermo-electric actuators includes multiple Nitinol strands, each Nitinol strand within a thermo-electric actuator of the plurality of thermo-electric actuators being independently controllable. In some examples, the prosthetic apparatus can further include a power supply connectable to the plurality of thermo-electric actuators.

[0007] Another aspect of the present disclosure relates to a thermo-electric actuator for prosthesis actuation. The thermo-electric actuator can include: a casing defining an internal volume; a first charge plate and a second charge plate positioned within the internal volume; a Nitinol band disposed within the internal volume and having opposing ends electrically coupled to the first charge plate and the second charge plate, respectively; and a movable shaft connected to the first charge plate, the movable shaft attachable to a cable.

[0008] In some examples, the Nitinol band has a thermal activation temperature ranging between 30 degrees Celsius and 80 degrees Celsius. In certain examples, the Nitinol band includes a helical coil positionally offset from the casing by a gap; and while maintaining the gap relative to the casing, the helical coil is configured to rotate and compress in response to thermal activation of the Nitinol band. In one or more examples, the casing includes a thermally conductive material configured to dissipate heat from the Nitinol band into an environment external to the casing.

[0009] In certain examples, the thermo-electric actuator can include a first insulator insulating the first charge plate from the casing; and a second insulator insulating the second charge plate from the casing. In some examples, the first charge plate and the first insulator define an opening; and the opening is configured to inhibit a pressure differential within the internal volume when the first charge plate and the first insulator is displaced relative to the second charge plate. In one or more examples, the second charge plate is configured to remain stationary within the casing. In particular examples, at least one of the first charge plate or the second charge plate is connectable to a power supply.

[0010] Yet another aspect of the present disclosure relates to a thermo-electric actuator for digit actuation of a prosthetic arm. The thermo-electric actuator can include: a casing defining an internal volume, the casing thermally coupling the internal volume to an environment external to the casing; a first charge plate and a second charge plate positioned opposite the first charge plate within the internal volume; a Nitinol helical coil having opposing ends electrically coupled to the first charge plate and the second charge plate, respectively; a movable shaft connected to the first charge plate, the movable shaft attachable to a cable for actuating a prosthetic digit; a seal positioned around the movable shaft; a first insulator insulating the first charge plate from the casing; a second insulator insulating the second charge plate from the casing; and a cap positioned around the second insulator, the second insulator being keyed to the second charge plate and the cap. In some examples, the first charge plate and the first insulator translate together toward the second charge plate when the Nitinol helical coil is heated above an activation temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

[0012] FIG. 1 illustrates a prosthetic limb in accordance with one or more examples of the present disclosure;

[0013] FIGS. 2A-2E illustrate prosthetic digit actuation utilizing a thermo-electric actuator in accordance with one or more examples of the present disclosure;

[0014] FIGS. 3-4 illustrate perspective views of a thermo-electric actuator in accordance with one or more examples of the present disclosure;

[0015] FIGS. 5A-5B illustrate exploded views of a thermo-electric actuator in accordance with one or more examples of the present disclosure;

[0016] FIG. 6 illustrates a profile view of a thermo-electric actuator in accordance with one or more examples of the present disclosure;

[0017] FIG. 7 illustrates a cross-sectional view of a thermo-electric actuator in accordance with one or more examples of the present disclosure; and

[0018] FIG. 8 illustrates another cross-sectional view of a thermo-electric actuator in accordance with one or more examples of the present disclosure.

DETAILED DESCRIPTION

[0019] Reference will now be made in detail to representative examples illustrated in the

accompanying drawings. It should be understood that the following descriptions are not intended to limit the examples to one preferred example. To the contrary, it is intended to cover alternatives, modifications, and equivalents as can be included within the spirit and scope of the described examples as defined by the appended claims.

[0020] The following disclosure relates to a lightweight prosthetic apparatus that implements one or more thermo-electric actuators. The prosthetic apparatus of the present disclosure can utilize low levels of energy (e.g., lower energy levels than some conventional prosthetic apparatuses) to efficiently power a thermo-electric actuator, resulting in a smaller and/or lighter power supply. With decreased weight, a prosthetic apparatus of the present disclosure can provide various advantages over conventional prosthetic arms. For example, the disclosed prosthetic apparatus can improve user ergonomics, decrease user fatigue, and increase fine motor control.

[0021] The thermo-electric actuator can be implemented in a variety of configurations. In some examples, the thermo-electric actuator includes a linear actuator. In particular examples, the thermo-electric actuator includes a shape memory material (e.g., a nickel titanium alloy known as “Nitinol”) activated by heat. In these or other examples, the thermo-electric actuator can include a pair of charge plates with at least one Nitinol strand electrically coupled to the pair of charge plates. When an electrical charge is selectively applied to the pair of charge plates, the Nitinol strand(s) disposed between the pair of charge plates act as resistance (and therefore build heat) that activates the Nitinol strand(s). Activation of the Nitinol strand(s) can actuate the thermo-electric actuator and a corresponding portion of the prosthetic, such as a prosthetic finger.

[0022] These and other examples are discussed below with reference to FIGS. **1-8**. However, a person of ordinary skill in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting. Furthermore, as used herein, a system, a method, an article, a component, a feature, or a sub-feature including at least one of a first option, a second option, or a third option should be understood as referring to a system, a method, an article, a component, a feature, or a sub-feature that can include one of each listed option (e.g., only one of the first option, only one of the second option, or only one of the third option), multiple of a single listed option (e.g., two or more of the first option), two options simultaneously (e.g., one of the first option and one of the second option), or combination thereof (e.g., two of the first option and one of the second option).

[0023] FIG. **1** illustrates a prosthetic arm **100** in accordance with one or more examples of the present disclosure. As shown, the prosthetic arm **100** can include a limb connection portion **102**. The limb connection portion **102** can be sized and shaped to attach to a user **114**. For example, the limb connection portion **102** can include a receptacle or socket sized and shaped to receive an end portion of a user's limb (e.g., an amputated limb, malformed limb, or missing limb). In some instances, the socket is a custom socket fabricated to intimately fit, engage, interlock, or mate with the residual limb of the user **114**. Additionally or alternatively, the limb connection portion **102** can include a compression sleeve, fastener, belt, sling, harness, etc. to secure the prosthetic arm **100** to the user **114**.

[0024] The prosthetic arm **100** can further include a hand portion **104** opposite the limb connection portion **102**. The hand portion **104** can include digits **106**. The digits **106** can include one or more of a finger, hook, grapple, grip, pincer, extension, or other limb-terminating device. In particular examples, the digits **106** can articulate or move (e.g., to bend, touch, pick up an object, squeeze, or point).

[0025] In these or other examples, the digits **106** can articulate in response to actuation of thermo-electric actuators **110**. As used herein, the term “thermo-electric actuator” refers to a variety of components, whether mechanical, electromechanical, hydraulic, pneumatic, piezoelectric, thermal, etc., that can generate controlled motion via energy and/or signals (e.g., electrical signals). The controlled motion can, in some examples, include rotary motion or linear motion. In certain examples, a thermo-electric actuator utilizes electrical resistance (particularly the thermal

byproduct of resistance) to drive a movable shaft. Additional detail of the thermo-electric actuators **110** is discussed further below in relation to FIGS. **3-8**.

[0026] The thermo-electric actuators **110** can be positioned at a variety of locations on or within the prosthetic arm **100**. In some examples, the thermo-electric actuators **110** are positioned between the hand portion **104** and the limb connection portion **102** (e.g., within the underside of the forearm region). In one or more examples, the thermo-electric actuators **110** are at least partially covered or encapsulated within a body or housing of the prosthetic arm **100** (e.g., to protect the thermo-electric actuators **110** from fluid ingress).

[0027] The prosthetic arm **100** can additionally include a power supply **108**. As used herein, the term “power supply” refers to any power source that supplies power to one or more components of the prosthetic arm **100** (e.g., to charge a battery and/or power the thermo-electric actuators **110**). For example, a power supply can include fuel cells, battery cells, generators, alternators, solar power converters, motion-based converters (e.g., that convert vibrations or oscillations into power), etc. In particular implementations, a power supply can convert alternating current to direct current (or vice-versa) for charging or recharging components of the prosthetic arm **100**. Some particular examples of a power supply can include a switched mode power supply, an uninterruptible power supply, an alternating current power supply, a direct current power supply, a regulated power supply, a programmable power supply, a computer power supply, and a linear power supply. In at least one example, a power supply includes a battery (e.g., a rechargeable battery) that can mount onto a battery mount. In a specific implementations, a power supply can include myoelectric technology that uses electrical signals generated by muscles of the user **114** (e.g., in the residual limb) to control movement of the prosthetic arm **100**. In one or more examples, the power supply **108** is integrated inside a body or housing of the prosthetic arm **100**. In other examples, the power supply **108** is exterior-integrated with the prosthetic arm **100**. In these or other examples, the power supply **108** can be electrically coupled with the thermo-electric actuators **110** to selectively actuate the thermo-electric actuators **110**.

[0028] The thermo-electric actuators **110** can be connected to the digits **106** via cables **112**. The cables **112** can include cable ropes, tensioned cable, wire, filament, string, tethers, linkages, pulleys, hinges, plungers, and/or other devices configured to transfer actuation of the thermo-electric actuators **110** to the digits **106**. Additionally or alternatively, the cables **112** can include actuatable shafts, toggles, latches, ratchets, clamps, friction brakes, toothed brakes, hydraulic brakes, non-back-drivable mechanisms, bearings/bushings, dovetail slides, scissor mechanisms, sarrus linkages, gears (e.g., worm gears, rotary gears), movable arms, winches, locking mechanisms, etc. In certain examples, the cables **112** can be implemented along a surface of the prosthetic arm **100**. In other examples, at least a portion of the cables **112** are positioned sub-surface and/or are at least partially concealed within the prosthetic arm **100**.

[0029] In particular examples, activation of the thermo-electric actuators **110** can cause the cables **112** to retract in and release out for desired digit actuation. For example, each cable of the cables **112** can move a respective digit of the digits **106** in response to actuation of a corresponding thermo-electric actuator **110**. For instance, the thermo-electric actuators **110** can retract or tension the cables **112** to cause the digits **106** to correspondingly retract inward (thereby inducing an anatomically-similar flexion of the digits **106** to curve or bend inward toward the palm of the hand portion **104** to a retracted position). Additional detail of example digit actuation or articulation is discussed further below in relation to FIGS. **2A-2E**.

[0030] It will be appreciated that the prosthetic arm **100** can include a controller to selectively release power from the power supply **108** to the thermo-electric actuators **110**. As used herein, the term “controller” can include a processor and a memory device. The processor can include at least one of a system on chip, integrated circuit, driver, microcontroller, application processor, crossover processor, etc. The memory device can include at least one of individual nonvolatile memory, processor-embedded nonvolatile memory, random access memory, memory integrated circuits,

DRAM chips, stacked memory modules, storage devices, memory partitions, etc. In these or other examples, the memory device can store computer-executable instructions that, when executed by the processor, cause the processor to perform certain steps, processes, etc. For instance, the processor can cause the power supply **108** to power one or more of the thermo-electric actuators **110**. In particular examples, the processor can control an amount and/or duration of current allowed from the power supply **108** to a given thermo-electric actuator (thereby facilitating fine motor control for enabling prolonged object holds, partial depressions or partial finger bends, etc.). That is, the processor can control the level of activation for the thermo-electric actuators **110** (and therefore the different retracted positions of the digits **106**) by throttling and/or increasing current flow from the power supply **108** to the thermo-electric actuators **110**. In specific implementations, the controller is communicatively coupled to one or more thermistors that are attached to, for example, Nitinol bands of the thermo-electric actuators **110**. In some examples, however, the controller is communicatively coupled to the Nitinol bands of the thermo-electric actuators **110** (where the Nitinol bands themselves can serve as thermistors having temperature readouts along one or more portions of the Nitinol bands).

[0031] The features of the prosthetic arm **100** can be modified in various ways. For example, although the prosthetic arm **100** is shown as implementing the thermo-electric actuators **110**, other types of prosthetic apparatuses can also implement the thermo-electric actuators **110**. To illustrate, other prosthetic apparatuses implementing the thermo-electric actuators **110** can include a prosthetic hand, foot, leg, shoulder, etc. As some specific examples, prosthetic apparatus implementing the thermo-electric actuators **110** can include a prosthetic apparatus for partial finger or partial thumb amputations, metacarpal or transmetacarpal amputations, wrist disarticulation, transradial amputation, elbow disarticulation, transhumeral amputation, shoulder disarticulation, forequarter amputation, etc. Accordingly, the prosthetic arm **100** is just one example implementation of a prosthetic apparatus implementing thermo-electric actuators.

[0032] As another example modification, the prosthetic arm **100** (or another prosthetic apparatus) can include motor-assist actuation. The motor-assist actuation can work in combination with (or independently from) the thermo-electric actuators **110**, as may be desired. Examples of a motor used for such motor-assist actuation can include an alternating current brushless motor, direct current (DC) brushed motor, DC brushless motor, direct drive motor, linear motor, servo motor, stepper motor, etc. Some specific examples of a DC motor include a DC shunt motor, separately excited motor, DC series motor, permanent magnet DC motor, DC compound motor, etc. Some specific examples of an AC motor include a synchronous motor or induction motor. Other examples of an electric motor include a reluctance motor, universal motor, or hysteresis motor.

[0033] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **1** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **1**.

[0034] FIGS. **2A-2E** illustrate prosthetic digit actuation utilizing a thermo-electric actuator in accordance with one or more examples of the present disclosure. A side profile view of a single digit **106** connected to the hand portion **104** is specifically depicted. As shown in FIG. **2A**, the digit **106** is in a rest position (also referred to as an extended position or an unactuated position). In the rest position of FIG. **2A**, the digit **106** can be substantially planar or parallel to the hand portion **104**. For example, a wrist joint **200**, a base knuckle joint **202**, a mid-knuckle joint **204**, a distal knuckle joint **206**, and a fingertip **210** can all be aligned relative to one another.

[0035] FIG. **2B** shows a first retracted position of the digit **106**. The first retracted position of the digit **106** can be achieved by causing the thermo-electric actuator **110** shown in FIG. **2A** to actuate

(or begin actuating), which in turn tensions or retracts the cable **112**. The cable **112** can, in some examples, extend through the wrist joint **200**, the base knuckle joint **202**, the mid-knuckle joint **204**, and the distal knuckle joint **206** before attaching to the fingertip **210** via a connection **208**. Thus, the fingertip **210** can begin to draw down and in toward an underside region **201** of the hand portion **104** as the thermo-electric actuators **110** is actuated (e.g., is thermally activated as will be described more below in relation to FIGS. 3-8).

[0036] FIG. 2C shows a second retracted position of the digit **106**. The second retracted position of the digit **106** can be achieved by causing the thermo-electric actuator **110** shown in FIG. 2B to further actuate, which in turn further tensions or retracts the cable **112**. Accordingly, the digit **106** in the second retracted position is—relative to the first retracted position—more curled or drawn inward closer to the palm of the hand portion **104**.

[0037] Although additional retracted positions are not shown, it will be appreciated that the digit **106** can be incrementally drawn inward to the underside region **201** of the hand portion **104** (e.g., to form a fist, enclose an object, etc.). In other words, the digit **106** can include a continuous range of adjustment resolution that is not bound to certain (e.g., predetermined) intermediate position settings. Of course, the term “continuous” should not be interpreted as necessarily requiring an infinite number of position settings because actuators do not typically have infinite adjustment resolution. Instead, actuators commonly actuate in a pseudo-continuous manner via finely-tuned increments (e.g., increments of micrometers, millimeters, centimeters, inches, etc.). The thermo-electric actuators **110** can also operate in a pseudo-continuous manner such that a magnitude of digit retraction to various retracted positions is dependent upon a level of thermal activation. For example, as the thermo-electric actuators **110** become increasingly thermally activated, the cable **112** can be correspondingly (e.g., proportionally) retracted.

[0038] Still, in other examples, the thermo-electric actuators **110** can be actuated in a step-wise fashion or between predetermined positions. For example, the thermo-electric actuators **110** can be actuated between two predetermined positions (e.g., based on Nitinol bands with two specific pre-trained shapes).

[0039] After actuated to the desired extent or the desired position, the thermo-electric actuators **110** can transition from an activated state to a resting state. The mechanics of this transition in relation to components of the thermo-electric actuators **110** are discussed further below in conjunction with the description of FIGS. 3-8. In general, however, the thermo-electric actuators **110** can transition from an activated state to a resting state by cooling down from a first temperature above an activation temperature (specific to the thermo-electric actuators **110**) to a second temperature below the activation temperature.

[0040] FIGS. 2D-2E show this incremental transition of the thermo-electric actuator **110** from an activated state (shown in FIGS. 2B-2C) to a resting state (depicted in FIG. 2E). For example, when the thermo-electric actuator **110** transitions from an activated state to a resting state, the digit **106** can move in an opposite direction (i.e., outward and away from the retracted position of FIG. 2C) to a resting position shown in FIG. 2E (where the digit **106** is substantially relaxed and aligned in-plane with the hand portion **104**).

[0041] To aid in moving the digit **106** from a retracted position to a resting position, certain embodiments of the prosthetic arm **100** can include a biasing structure **212**. The biasing structure **212** can include a spring, elastic band, etc. that can store potential energy as the digits **106** moves inward responsive to actuation of the thermo-electric actuator **110**. Then, as the thermo-electric actuator **110** transitions to a resting state, the biasing structure **212** can use the stored potential energy (e.g., in the form of an induced tensile force in the biasing structure **212**) to bias or pull the digit **106** back to a resting position. Functionally, in some examples, the biasing structure **212** can provide a spring-back motion for the digit **106** once the thermo-electric actuator **110** is no longer activated. Additionally or alternatively, the biasing structure **212** can help ensure the digit **106** follows a predetermined let-down or path of travel from a retracted position to a resting position.

[0042] The biasing structure **212** can be mounted to the prosthetic arm **100** in various locations and orientations. In some examples, the biasing structure **212** is positioned on top of or at least partially embedded within a top portion of the hand portion **104** (e.g., to impart the outward rotational movement for the digit **106**). In particular examples, the biasing structure **212** includes an end portion attached to a connection **214** mounted between the base knuckle joint **202** and the wrist joint **200**. The biasing structure **212** can include an opposing end portion attached to a connection **216** mounted between the mid-finger knuckle joint **204** and the fingertip **210**.

[0043] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 2A-2E can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 2A-2E.

[0044] FIGS. 3-4 illustrate perspective views of the thermo-electric actuator **110** in accordance with one or more examples of the present disclosure. As shown, the thermo-electric actuator **110** can include a casing **300**, a movable shaft **302** extending from a first end **304** opposite a second end **306**, and a cap **308**.

[0045] In more detail, the casing **300** can include a shell, cover, enclosure, frame, sleeve, or collar forming a body portion of the thermo-electric actuator **110**. In some examples, the casing **300** thermally couples the internal volume and an environment external to the thermo-electric actuator **110**. For instance, the casing **300** can include a thermally conductive material (e.g., aluminum, copper, gold, silver, bismuth, graphene, carbon nanotube, diamond, combinations thereof, etc.) to dissipate heat from components within the internal volume to the environment external to the casing **300**. In specific implementations, the casing **300** can also be electrically insulative (e.g., aluminium nitride). For example, the casing **300** can be coated with an electrical insulator, such as a ceramic material.

[0046] The casing **300** can also define an internal volume of the thermo-electric actuator **110** where inner components of the thermo-electric actuator **110** can be housed. The casing **300** can therefore include a variety of configurations for an inner housing, capsule, hull, pod, chamber, etc. As shown in other figures, however, the casing **300** can define a cylindrically-shaped internal volume for active displacement of various components of the thermo-electric actuator **110**.

[0047] As an example, the movable shaft **302** at the first end **304** can move in and out as thermo-electric actuator **110** is actuated (e.g., thermally activated). The movable shaft **302** can extend through a central opening in the casing **300** and attach to a cable (e.g., one of the cables **112** discussed above) for actuating a prosthetic apparatus. Examples of the movable shaft **302** can include a piston, rod, shank, dowel, etc. that moves in tandem with internal components of the thermo-electric actuator **110** (discussed more below). For instance, as the movable shaft **302** retracts inward (i.e., into the casing **300**) in response to thermal activation of the thermo-electric actuator **110**, the movable shaft **302** can correspondingly pull or tension a cable attached to the movable shaft **302**.

[0048] With respect to FIG. 4, the thermo-electric actuator **110** can include the cap **308** on the second end **306**. In these or other examples, the cap **308** can include a lid, cover, base, or locking mechanism positioned on the second end **306**. In some examples, the cap **308** can enclose a bottom end of the casing **300**. Additionally or alternatively, the cap **308** can at least partially surround other components of the second end **306**. For example, the cap **308** can be positioned around a charge plate **400** and an insulator **402**. In some implementations, the cap **308** is integrally formed with the casing **300** at the second end **306**. In other implementations, the cap **308** is attached (e.g., adhered or fastened) to the casing **300** at the second end **306**.

[0049] The charge plate **400** can include a stationary base electrically coupled to one or more

components within the internal volume of the thermo-electric actuator **110**. Additionally or alternatively, the charge plate **400** can be electrically coupled to one or more components external to the thermo-electric actuator **110**. For example, the charge plate **400** can be connected to a negative lead on a power supply (e.g., the power supply **108** discussed above). As another example, the charge plate **400** can be connected to an electrical ground (e.g., a chassis or body portion of the prosthetic arm **100**). As will be discussed more below, the charge plate **400** can be oppositely charged relative to another charge plate disposed inside the thermo-electric actuator **110** (e.g., the charge plate **502** discussed below in relation to at least FIG. 5). Accordingly, the charge plate **400** can include an electrically conductive material (e.g., copper, aluminum, gold, silver, steel, platinum, zinc, graphite, nickel, brass, tungsten, combinations thereof, etc.).

[0050] In one or more examples, the insulator **402** can electrically insulate the charge plate **400** from the casing **300**. Electrically insulating the charge plate **400** from the casing **300** can help maintain a desired charge for the charge plate **400** (thereby helping to control thermal activation of the thermo-electric actuator **110**). In some examples, the insulator **402** can include a polymer material, composite material, glass material, porcelain material, rubber material, etc.

[0051] In specific implementations, one or more components of the second end **306** can interlock with or be keyed to one another. For example, the insulator **402** can be keyed to the charge plate **400** and the cap **308**. Additionally or alternatively, the components of the second end **306** can be fastened or bonded together. In this manner, the second end **306** can secure various components of the thermo-electric actuator **110** assembled together and inhibit component creep to maintain relative component spacing.

[0052] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 3-4 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 3-4.

[0053] FIGS. 5A-5B illustrates an exploded view of the thermo-electric actuator **110** in accordance with one or more examples of the present disclosure. The thermo-electric actuator **110** can include at least one Nitinol band **500** (or as shown, multiple Nitinol bands **500** spaced apart). The terms “band” and “strand” in conjunction with Nitinol refer to an elongate segment, continuous portion, weaving, filament, fibril, thread, coil, length, mesh, hollow tube, or constituent piece of Nitinol material. A band or strand of Nitinol can have a myriad of different shapes, cross-sectional profiles, and sizes. In some examples, a Nitinol band can include a single filament of Nitinol. In other examples, a Nitinol band can include a combination of multiple Nitinol bands integrally joined (e.g., fused together in a predetermined shape, mesh configuration, tube, linked arrangement, etc.). Still, in other examples, a Nitinol band can include a combination of multiple, discrete Nitinol bands positioned adjacent to each other in intimate contact (e.g., intertwined, twisted, or braided together).

[0054] In these or other examples, the Nitinol bands **500** can include various alloys of nickel and titanium. In particular implementations, the Nitinol bands **500** can include a ratio of about 50 atomic percent for each of nickel and titanium (or about 55 percent by weight of nickel). The foregoing ratios, however, can be modified. For instance, the ratio of nickel and titanium can be adjusted to change the austenite yield strength, the activation temperature at which the Nitinol bands **500** can transition to a trained shape (or active state), etc. Additives can also be implemented to tune various properties of the alloy.

[0055] In certain examples, the Nitinol bands **500** can include a variety of activation temperatures (e.g., ranging between 30 degrees Celsius and 80 degrees Celsius). The term “activation temperature” refers to the temperature of Nitinol material at which the Nitinol begins to transition

between the Martensitic phase and the Austenite phase. As the activation temperature is increased, the time to activation can correspondingly increase (lending to slightly slower response times to retract one or more digits of the prosthetic arm **100**). However, with higher activation temperatures, the cooling time to drop below the activation temperature can be correspondingly decreased (lending to slightly faster response times to relax or release one or more digits of the prosthetic arm **100** from a retracted position to a resting position). The converse is also true. As the activation temperature is decreased, the time to activation can correspondingly decrease (lending to slightly faster response times to retract one or more digits of the prosthetic arm **100**). Thus, with a lower activation temperature, the cooling time to drop below the activation temperature can be correspondingly increased (lending to slightly slower response times to relax or release one or more digits of the prosthetic arm **100** from a retracted position to a resting position). It will be appreciated, therefore, that the activation temperature of the Nitinol bands **500** can be tuned and adjusted (or customized) for desired digit actuation response rates. In some examples, digit actuation response rates can range from microseconds to seconds.

[0056] As mentioned above, the Nitinol bands **500** can include a trained shape such that, upon heating the Nitinol bands **500** above an activation temperature, the Nitinol bands **500** can recover (i.e., move or deform) to the trained shape from another shape (e.g., a non-activated shape, a resting shape, a non-contracted shape, a lengthened shape, a deformed shape, etc.). The trained shape can include any shape, configuration, and/or orientation of the Nitinol bands **500** that occurs above the activation temperature. In specific implementations, the trained shape corresponds to a final shape associated with the Active Austenite Finish (Active Af) temperature of the Nitinol at which the shape recovery is complete. However, it will be appreciated that achieving the trained shape is an incremental process and the Nitinol bands **500** can therefore achieve a portion of the trained shape when a temperature of the Nitinol bands **500** exceeds the activation temperature and progressively warms. The many trainable shapes of the Nitinol bands **500** are herein contemplated. However, in specific implementations of the thermo-electric actuator **110**, a trainable shape of the Nitinol bands **500** can include a compressed or contracted helical coil.

[0057] Further, in some examples, the Nitinol bands **500** can include multiple trained shapes (e.g., a first trained shape for retracting the thermo-electric actuator **110** and a second trained shape for protracting the thermo-electric actuator **110**). In such an example, the biasing structure **212** (or other return mechanism) discussed above may be omitted in lieu of a multiple trainable shapes for the Nitinol bands **500**.

[0058] In these or other examples, the Nitinol bands **500** can include opposing ends electrically coupled to respective charge plates, namely the charge plate **400** (discussed above) and a charge plate **502** disposed within an internal volume **510** of the thermo-electric actuator **110**. The charge plate **502** can be the same as or similar to the charge plate **400**. Via the charge plates **400**, **502**, the Nitinol bands **500** can be thermally activated. For example, the charge plate **502** can be electrically coupled to a positive lead on a power supply (e.g., the power supply **108**), and the charge plate **400** can be electrically coupled to a negative lead on a power supply (or an electrical ground). The Nitinol bands **500** can therefore act as resistance in response to an applied or induced charge at the charge plates **400**, **502**. In turn, the Nitinol bands **500** can be heated to actuate the thermo-electric actuator **110**. Specifically, as the Nitinol bands **500** heats above the activation temperature, the Nitinol bands **500** can draw down the charge plate **502** and the movable shaft **302** along axis **508**. That is, at least these components can actuate together as a connected system because the Nitinol bands **500** is attached to the charge plate **502**, and the charge plate **502** is integrally connected to or otherwise attached to the movable shaft **302**.

[0059] In some examples, each Nitinol band or strand within the thermo-electric actuator **110** can be independently heat controlled. For example, each end of each Nitinol band can be electrically connected to its own (separate and independent) power connection on the charge plates **400**, **502**. In this manner, each of the Nitinol bands **500** can be precisely controlled for a more consistent

thermal activation (lending to a reduced temperature discrepancy between the Nitinol bands **500** and therefore a more efficient actuating motion for the thermo-electric actuator **110**). In these or other examples, each individual band of the bands **500** can have a thermistor connected to the band for precise temperature monitoring and control over the overall actuator contraction rate and the position of the movable shaft **302**. Alternatively, in some examples, each individual band of the bands **500** can act as a thermistor itself, having temperature readouts along one or more portions of the Nitinol bands (e.g., for comparing temperatures across different band portions or between different bands).

[0060] The thermo-electric actuator **110** can further include an insulator **504**. The insulator **504** can electrically insulate the charge plate **502** (e.g., from contacting the casing **300**). In these or other examples, the insulator **504** can be positioned on top of the charge plate **502** (e.g., to insulate a top surface of the charge plate **502**). Additionally, in some examples, the insulator **504** can be positioned around the sides of the charge plate **502** (e.g., to insulate the sidewall of the charge plate **502**). In particular implementations, the insulator **504** is adhered, fastened, or otherwise attached to the charge plate **502**. The openings of the insulator **504** can correspond to openings in the charge plate **502** (as shown and described in relation to FIG. **8** to inhibit a pressure differential within the internal volume **510** of the thermo-electric actuator **110**).

[0061] The thermo-electric actuator **110** can additionally include a seal **506** (e.g., a gasket). The seal **506** can be disposed around the movable shaft **302** between the casing **300** and the insulator **504**. In some examples, the seal **506** can maintain engagement with the movable shaft **302** as the movable shaft **302** translates in and out of the thermo-electric actuator **110**. In so doing, the seal **506** can seal out air, particles, fluids, etc. from being introduced into the internal volume **510**.

[0062] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. **5A-5B** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. **5A-5B**.

[0063] FIG. **6** illustrates a profile view of a thermo-electric actuator in accordance with one or more examples of the present disclosure. In particular, FIG. **6** depicts the thermo-electric actuator **110** as having a length **600**. The length **600** can vary depending on the type of prosthetic apparatus or desired actuation. For example, the length **600** can be comparatively larger for actuation of an entire limb (e.g., an arm or leg) than for actuation of digits (e.g., fingers or toes). In particular examples, however, the length **600** can range from about 1 inch to about 8 inches, about 2 inches to about 6 inches, or about 3 inches to about 5 inches.

[0064] As mentioned above, the movable shaft **302** can actuate in and out of the thermo-electric actuator **110** to correspondingly draw in and release a cable attached to the movable shaft **302**. FIG. **6** specifically depicts an actuation direction **602** for the movable shaft **302**. That is, as the Nitinol bands **500** are heated above an activation temperature, the movable shaft **302** is drawn into the thermo-electric actuator **110** along the actuation direction **602**.

[0065] A displacement distance **604** for the movable shaft **302** can correspond to the magnitude of digit retraction shown in FIGS. **2B-2C**. Similarly, the displacement distance **604** can correspond to the level of thermal activation of the Nitinol bands **500**. For instance, the displacement distance **604** can increase as the temperature of the Nitinol bands **500** increases beyond the activation temperature and approaches the Active Af temperature of the Nitinol at which the shape recovery is complete.

[0066] The displacement distance **604**, like the length **600**, can depend on the application of the thermo-electric actuator **110**. For example, the displacement distance **604** can be larger for actuating an entire limb (e.g., an arm or a leg) versus actuating a digit (e.g., a finger or toe). In

particular implementations (e.g., for actuating a digit), the displacement distance **604** can range from about a half centimeter to about 10 centimeters, about 1 centimeter to about 8 centimeters, about 2 centimeters to about 6 centimeters, or about 3 centimeters to about 4 centimeters.

[0067] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIG. **6** can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIG. **6**.

[0068] FIGS. **7-8** illustrate cross-sectional views of the thermo-electric actuator **110** in accordance with one or more examples of the present disclosure. In particular, FIG. **7** shows the Nitinol bands **500** disposed within the internal volume **510** and connected to the charge plate **400** and the charge plate **502**. As mentioned above, the Nitinol bands **500** can contract in response to an applied charge to the charge plate **400** and the charge plate **502** heating the Nitinol bands **500** above an activation temperature. As the Nitinol bands **500** contracts, each of the charge plate **502**, the insulator **504**, and the movable shaft **302** can translate together toward the charge plate **400** (i.e., in the actuation direction **602** along the axis **508**). As this actuation occurs, the charge plate **400** can remain stationary relative to the casing **300** because the charge plate **400** is attached or adhered to the insulator **402** (which is also stationary and can interlock with or otherwise attach to the cap **308**). Additionally, in some examples, the Nitinol bands **500** (and therefore the charge plate **502**, the insulator **504**, and the movable shaft **302**) can rotate about the axis **508** while translating during actuation.

[0069] Further, as the Nitinol bands **500** are activated and actuation begins to occur, a spacing **700** between the Nitinol bands **500** can change. For example, the spacing **700** can decrease during as the Nitinol bands **500** are activated and move to a trained shape (e.g., a contracted or compressed helical coil). Conversely, the spacing **700** can increase as the Nitinol bands **500** transition to a resting state and the charge plate **502** translates away from the charge plate **400**. In some examples, the spacing **700** is always greater than zero such that the Nitinol bands **500** do not contact each other. In other examples, the spacing **700** can equal zero when the Nitinol bands **500** are activated such that one or more of the Nitinol bands **500** can contact each other (or in the case of a single strand, adjacent coils of the single strand can be compressed into contact with each other).

[0070] In these or other examples, the Nitinol bands **500** can maintain a gap **702** relative to the casing **300**. In this manner, the Nitinol bands **500** is allowed to flex or move in transitioning to a trained shape, while still avoiding contact with the casing **300**. In this manner, the casing **300** is prevented from electrically grounding one or more portions of the Nitinol bands **500**. Additionally or alternatively, the gap **702** can help reduce friction (e.g., by preventing friction between the Nitinol bands **500** and the casing **300**) to facilitate more efficient actuation of the thermo-electric actuator **110**. In some examples, the gap **702** can range from about 0.01 inches to about 0.5 inches, about 0.02 inches to about 0.1 inches, about 0.03 inches to about 0.08 inches, or about 0.05 inches to about 0.07 inches. The gap **702** can also be decreased, for example, due to a coating disposed along the interior wall of the casing **300** (e.g., an electrical insulator coating such as a ceramic coating).

[0071] As mentioned above, the casing **300** can dissipate heat from the Nitinol bands **500** into an environment external to the casing **300**. In some examples, internal volume **510**, can include additional elements to aid heat dissipation. For example, the internal volume **510** can include one or more non-charged fluids to cool the Nitinol bands **500**. As some specific examples, a non-charged fluid can include deionized water, oil, air, etc. In such examples, the internal volume **510** can function as a closed-system. In other examples, however, the internal volume **510** can be an open system (e.g., subject to ventilation via fans, pumps, etc. using air from the external

environment to actively cool the Nitinol bands 500).

[0072] FIG. 8 shows a cross-sectional view of a top portion of the thermo-electric actuator 110 where the Nitinol bands 500 are attached to the charge plate 502. In particular, this figure depicts openings 800 defined by the charge plate 502 and the insulator 504. The openings 800 can inhibit a pressure differential within the internal volume 510 when the charge plate 502 and the insulator 504 are displaced relative to the charge plate 400. In some examples, reducing or eliminating a pressure differential can facilitate a more efficient actuation of the thermo-electric actuator 110 such that displacement of charge plate 502 and the insulator 504 are not hindered.

[0073] Any of the features, components, and/or parts, including the arrangements and configurations thereof shown in FIGS. 7-8 can be included, either alone or in any combination, in any of the other examples of devices, features, components, and parts shown in the other figures described herein. Likewise, any of the features, components, and/or parts, including the arrangements and configurations thereof shown and described with reference to the other FIGS. can be included, either alone or in any combination, in the example of the devices, features, components, and parts shown in FIGS. 7-8.

[0074] The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described examples. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described examples. Thus, the foregoing descriptions of the specific examples described herein are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the examples to the precise forms disclosed.

[0075] It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings. Indeed, various inventions have been described herein with reference to certain specific aspects and examples. However, they will be recognized by those skilled in the art that many variations are possible without departing from the scope and spirit of the inventions disclosed herein. Specifically, those inventions set forth in the claims below are intended to cover all variations and modifications of the inventions disclosed without departing from the spirit of the inventions. The terms “including” or “includes” as used in the specification shall have the same meaning as the term “comprising.” The terms “about” or “approximately” should be interpreted as ± 10 percent of a given value, unless otherwise indicated.

Claims

1. A prosthetic apparatus, comprising: a limb connection portion attachable to a user; a hand portion opposite the limb connection portion, the hand portion comprising a plurality of digits; and a plurality of thermo-electric actuators positioned between the hand portion and the limb connection portion, the plurality of thermo-electric actuators connected to the plurality of digits.
2. The prosthetic apparatus of claim 1, further comprising a plurality of cables connecting the plurality of thermo-electric actuators to the plurality of digits, wherein upon thermal activation of a thermo-electric actuator of the plurality of thermo-electric actuators, the thermo-electric actuator retracts a cable of the plurality of cables to move a digit of the plurality of digits inward to a retracted position.
3. The prosthetic apparatus of claim 2, wherein a magnitude of digit retraction to the retracted position is dependent upon a level of the thermal activation.
4. The prosthetic apparatus of claim 2, wherein the thermo-electric actuator of the plurality of thermo-electric actuators is configured to transition from an activated state to a resting state by cooling down from a first temperature above an activation temperature to a second temperature below the activation temperature.
5. The prosthetic apparatus of claim 4, wherein upon transitioning the thermo-electric actuator of the plurality of thermo-electric actuators from the activated state to the resting state, the digit of the

plurality of digits is configured to extend outward and away from the retracted position to a resting position.

6. The prosthetic apparatus of claim 5, further comprising a biasing structure to bias the digit of the plurality of digits toward the resting position.

7. The prosthetic apparatus of claim 6, wherein: a first end of the biasing structure is connected to a top portion of the digit between a mid-finger knuckle joint and a fingertip; and a second end of the biasing structure is connected to a top portion of the hand portion between a base knuckle joint and a wrist joint.

8. The prosthetic apparatus of claim 1, wherein each thermo-electric actuator of the plurality of thermo-electric actuators comprises at least one Nitinol strand.

9. The prosthetic apparatus of claim 8, wherein each thermo-electric actuator of the plurality of thermo-electric actuators comprises multiple Nitinol strands, each Nitinol strand within a thermo-electric actuator of the plurality of thermo-electric actuators being independently controllable.

10. The prosthetic apparatus of claim 1, further comprising a power supply connectable to the plurality of thermo-electric actuators.

11. A thermo-electric actuator for prosthesis actuation, the thermo-electric actuator comprising: a casing defining an internal volume; a first charge plate and a second charge plate positioned within the internal volume; a Nitinol band disposed within the internal volume and having opposing ends electrically coupled to the first charge plate and the second charge plate, respectively; and a movable shaft connected to the first charge plate, the movable shaft attachable to a cable.

12. The thermo-electric actuator of claim 11, wherein the Nitinol band has a thermal activation temperature ranging between 30 degrees Celsius and 80 degrees Celsius.

13. The thermo-electric actuator of claim 11, wherein: the Nitinol band comprises a helical coil positionally offset from the casing by a gap; and while maintaining the gap relative to the casing, the helical coil is configured to rotate and compress in response to thermal activation of the Nitinol band.

14. The thermo-electric actuator of claim 11, wherein the casing comprises a thermally conductive material configured to dissipate heat from the Nitinol band into an environment external to the casing.

15. The thermo-electric actuator of claim 11, further comprising: a first insulator insulating the first charge plate from the casing; and a second insulator insulating the second charge plate from the casing.

16. The thermo-electric actuator of claim 15, wherein: the first charge plate and the first insulator define an opening; and the opening is configured to inhibit a pressure differential within the internal volume when the first charge plate and the first insulator is displaced relative to the second charge plate.

17. The thermo-electric actuator of claim 11, wherein the second charge plate is configured to remain stationary within the casing.

18. The thermo-electric actuator of claim 11, wherein at least one of the first charge plate or the second charge plate is connectable to a power supply.

19. A thermo-electric actuator for digit actuation of a prosthetic arm, the thermo-electric actuator comprising: a casing defining an internal volume, the casing thermally coupling the internal volume to an environment external to the casing; a first charge plate and a second charge plate positioned opposite the first charge plate within the internal volume; a Nitinol helical coil having opposing ends electrically coupled to the first charge plate and the second charge plate, respectively; a movable shaft connected to the first charge plate, the movable shaft attachable to a cable for actuating a prosthetic digit; a seal positioned around the movable shaft; a first insulator insulating the first charge plate from the casing; a second insulator insulating the second charge plate from the casing; and a cap positioned around the second insulator, the second insulator being keyed to the second charge plate and the cap.

20. The thermo-electric actuator of claim 19, wherein the first charge plate and the first insulator translate together toward the second charge plate when the Nitinol helical coil is heated above an activation temperature.
